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Local Pluralistic Homophily for Boundary-Spanning Node Detection in Overlapping Community Networks

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We introduce *local pluralistic homophily* ($\tilde{h}_v$), a neighborhood–centered metric to identify inter–community bridges in overlapping networks. Large negative $\tilde{h}_v$ marks nodes whose membership profiles misalign with those of their neighbors, indicating boundary–spanning positions. We evaluate $\tilde{h}_v$ via targeted node removal across eight real–world networks, comparing against both topology–based baselines and a recent community–aware alternative. Two axes assess boundary effects: a mechanistic axis quantifying reductions in cross–community coupling, and a functional axis measuring preferential targeting of community interfaces. Removing low–$\tilde{h}_v$ nodes depletes inter–community coupling more rapidly than random while preserving global connectivity, and achieves higher interface selectivity than both baseline families in most networks. Rank correlations with structural measures are predominantly negative, showing that $\tilde{h}_v$ surfaces membership–based boundaries rather than structural hubs. As an unsupervised, overlap–based, and computationally lightweight method (near–linear time), $\tilde{h}_v$ offers a practical basis for auditing cross–group information flow, stress–testing community resilience, and prioritizing boundary–spanning interventions.

1. Introduction

Identifying boundary–spanning nodes is central to analyzing and understanding social, information, and biological networks. Boundary nodes mediate exposure, facilitate diffusion between groups, and shape segregation or integration dynamics [1]. While centrality measures have been widely used to identify influential nodes [2,3], established approaches for boundary detection—betweenness centrality [4] and bridging centrality [5]—operate primarily on network topology, identifying nodes that broker connectivity through sparse local structures or positions along shortest paths.

Notably, many real–world networks exhibit overlapping community structure, where nodes participate in multiple groups simultaneously [6]. In such settings, boundary positions can emerge not only from sparse connectivity but also from *membership misalignment*: nodes whose community–membership profiles differ from those of their neighbors, thereby bridging groups with distinct participation patterns. Topology–based methods do not fully leverage membership information: a node can span communities without low clustering or high shortest–path centrality, and conversely, a structural bridge may reside entirely within a single dense community [7]. This gap motivates a complementary approach based on membership–neighborhood alignment rather than topological properties alone.

We introduce a membership–based boundary score, the local pluralistic homophily coefficient $\tilde{h}_v$, which quantifies how well the community memberships of a node align

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with those of its neighbors in overlapping networks, extending pluralistic homophily [8] from a network–level coefficient [9] to node–level rankings for boundary detection. Three properties distinguish our approach. First, $\tilde{h}_v$ acts orthogonally to established baselines: it prioritizes membership–neighborhood misalignment rather than structural sparsity (BridgeCC [5,7]) or intra-community density (mDC [10]), yielding rankings that diverge from both purely topological and community-embeddedness measures. Second, it is computationally efficient: in sparse graphs with modest overlap, the end–to–end cost is near–linear in the number of edges. Third, it is interpretable: large negative values indicate membership disassortativity characteristic of nodes that connect communities with dissimilar participation profiles.

We evaluate whether membership–based boundaries form a functionally distinct population using targeted node–removal experiments. Nodes are removed in ascending $\tilde{h}_v$ (lowest first) and outcomes are compared against topology–based baselines and random removal at matched budgets. We assess two axes: (i) mechanistic depletion of inter–community coupling and (ii) functional selectivity for community interfaces, comparing against both topology–based (BridgeCC) and community–aware (mDC) baselines. As a connectivity control, we measure the giant–component fraction to ensure observed effects reflect boundary targeting rather than premature fragmentation.

We evaluate our approach across eight real-world networks spanning diverse domains—including collaboration networks (DBLP), social platforms (LiveJournal, YouTube), and content–interaction systems (GitHub, StackOverflow)—with ground–truth or metadata–derived community assignments. We find that membership–based boundaries constitute a functionally distinct population: removing nodes in ascending $\tilde{h}_v$ order depletes inter–community coupling more rapidly than random removal while preserving large–scale connectivity. Rank correlations reveal that low–$\tilde{h}_v$ nodes occupy positions distinct from both structural hubs (predominantly negative correlations with BridgeCC) and community–embedded nodes (intermediate correlations with mDC). This orthogonality translates into practical advantages: at matched budgets, $\tilde{h}_v$ achieves superior interface selectivity compared to topology–based baselines in seven of eight networks and outperforms the community–aware baseline mDC in six of eight, with variation reflecting differences in community formation mechanisms and overlap structure.

Our contributions are threefold. First, we introduce a membership–based boundary detection method that is unsupervised, computationally efficient, and tailored to overlapping communities. Second, we establish that membership–neighborhood misalignment offers a structural perspective orthogonal to both topology–based centrality and community–density measures: membership–based boundaries occupy positions distinct from structural hubs and community–embedded nodes, capturing nodes whose influence arises from cross–group membership bridging rather than connectivity brokerage or intra–community cohesion. Third, we show that this complementary perspective enables selective intervention: targeting membership misalignment reduces cross–group exposure while preserving within–group cohesion, a capability not fully achieved through either topology–based or community–embeddedness targeting alone [11].

The remainder of the paper is organized as follows. Section 2 reviews assortativity, overlapping communities, and related work on boundary detection. Section 3 defines the local pluralistic homophily score, details datasets, and describes the node–removal experiment design. Section 4 presents structural characterization and outcomes. Section 5 discusses implications, limitations, and future directions.

2. Theoretical Background

2.1. Assortativity and Pluralistic Homophily

Assortativity quantifies the tendency of connected nodes to share similar attributes [12]. Positive values indicate assortative mixing, where similar nodes preferentially connect; negative values indicate disassortativity, where dissimilar nodes connect; values near zero suggest random mixing. Empirically, social networks often exhibit positive assortativity, while biological and some informational networks show disassortative patterns [13,14]. Mixing can be assessed at the network level via correlation measures or at the node level via local variants that probe neighborhood-specific patterns [15,16].

We model a network as an undirected, unweighted graph $G = (V, E)$, where $N(v) = \{u \in V : (v, u) \in E\}$ denotes the neighborhood of node v. A *community* is a subset $c \subseteq V$ exhibiting higher internal connectivity than external [17,18]. Let $\mathcal{C} = \{c_1, \ldots, c_n\}$ denote the set of communities. In many real systems, communities overlap and nodes participate in multiple groups simultaneously [6]. For each node v, let $\mathcal{C}(v) = \{c \in \mathcal{C} : v \in c\}$ denote its community memberships and $s_v = |\mathcal{C}(v)|$ its *membership count*.

Pluralistic homophily, introduced by [8], captures the tendency of nodes to connect based on the number of communities they share. Barraza et al. [9] formalized this concept at the network level as an assortativity-style coefficient, using the number of communities per node as the attribute to quantify correlation in membership counts across edges. Let e_{xy} be the joint probability that a uniformly sampled edge connects nodes with membership counts x and y, and let q_x be the corresponding marginal (equivalently, the degree-weighted node distribution). With μ_q and σ_q^2 denoting the mean and variance of q_x, the network-level pluralistic homophily is:

$$h = \frac{\sum_{x,y}(x - \mu_q)(y - \mu_q)\,(e_{xy} - q_x q_y)}{\sigma_q^2}. \tag{1}$$

This network-level metric does not provide node-level rankings for boundary detection. A local decomposition $h = \sum_{v \in V} h_v$, analogous to local assortativity [15], computes the contribution of each node:

$$h_v = \frac{1}{2M\,\sigma_q^2} \sum_{i \in N(v)} (s_v - \mu_q)(s_i - \mu_q), \tag{2}$$

where $M = |E|$ is the number of edges. By construction, $\sum_{v \in V} h_v = h$ maintains consistency between node-level and network-level perspectives. However, unnormalized contributions h_v are not directly comparable for ranking purposes due to degree and variance heterogeneity.

2.2. Related Work

Identifying nodes whose removal weakens inter–community connectivity traces back to the theory of structural holes [1]. Classical approaches detect bridge or broker nodes using flow- and centrality-based heuristics [19], with methods such as BridgeCC emphasizing low clustering around moderately high-degree vertices. Recent work continues to refine boundary and brokerage concepts through new diagnostics and empirical analyses [20–22].

Community-aware heuristics explicitly leverage overlap structure. Degree disassortativity and membership mismatch frequently emerge in large networks [23], motivating rules that couple degree information with overlap cues. Examples include overlap-seed propagation [24], multi-community neighborhood rules [25], and the identification of highly connected overlap hubs [26]. Recent formulations integrate disassortativity with

community embeddedness, as in DoN/mDC [10]. Our formulation differs by measuring membership–neighborhood misalignment rather than degree contrast or local density.

A complementary line of research focuses on discovering overlapping communities rather than assessing boundaries within them. Overlapping detection methods often rely on seed expansion guided by influence or local structural patterns. Representative examples include the KNN-enhanced NOCD algorithm [27], influential-node identification for community expansion [28], maximum-entropy random walks [29], and higher-order, motif-based expansion [30]. These methods aim to reconstruct the community structure; in contrast, our approach assumes communities as given—whether ground truth or algorithmically detected—and identifies boundary nodes based on membership–neighborhood alignment.

Supervised approaches constitute another family, where bridge or boundary detection is framed as a learning task. For instance, BND [31] trains GNN models to label bridge nodes using topology and features, while recent work incorporates temporal dynamics to detect evolving structural-hole positions [32]. Such pipelines can be powerful when labels or rich features exist but require training data, hyperparameter tuning, and often presuppose community partitions with limited overlap.

At the application level, boundary-spanning nodes play key roles in shaping cross-group exposure and echo-chamber effects [33,34]. They are central to targeted interventions in epidemic control [35,36], network resilience [37,38], influence maximization [39], immunization strategies [40], and control in community-structured networks [41]. Community-aware centralities exhibit distinct performance depending on overlap and spreading regimes [11].

In summary, existing methods variously rely on topology, degree contrast, supervised learning, or community-detection pipelines. Our contribution is orthogonal to these families: we introduce a membership-based characterization of neighborhood misalignment that (i) captures boundary interfaces even when not topologically sparse, (ii) embraces overlap by design, and (iii) remains unsupervised, parameter-light, and near-linear in complexity.

3. Materials and Methods

We define the node–level pluralistic homophily score $\tilde{h}_v$ for boundary identification in overlapping networks and describe datasets, two baselines—a topology–based comparator (BridgeCC) and a community–aware baseline (mDC)—and the evaluation protocol. Two assessments are used: depletion of inter–community coupling under targeted removal and interface selectivity measured by the inter–community edge ratio (IER), where $\tilde{h}_v$ is compared against both baselines.

3.1. Local Pluralistic Homophily Score for Boundary Detection

The network-level pluralistic homophily coefficient h (Eq. 1) quantifies correlation in membership-related attributes across edges but does not provide node-level rankings for boundary detection. A local decomposition $h = \sum_{v \in V} h_v$ exists, where the contribution of each node is given by Eq. 2. However, as shown for degree assortativity [42], using a network-wide mean as a reference can mask neighborhood-specific variation. In particular, the magnitudes of h_v are not directly comparable for ranking because degree and variance heterogeneity can amplify values independently of the boundary role of a node [43].

To illustrate, consider a node v with a high membership profile connected to neighbors with similarly high profiles. Even if v is locally assortative, a low network-wide average makes $(s_v - \mu_q)$ large and dominates h_v. Conversely, a boundary node whose membership

profile differs substantially from its neighbors may exhibit a modest h_v in a low-variance region. Global centering can therefore mask neighborhood-local boundary patterns.

These limitations of h_v suggest that boundary detection requires a different underlying pattern. A boundary node is not characterized by its deviation from a *global* membership baseline, but by how its membership profile contrasts with those present in its *local neighborhood*. Conceptually, boundary positions arise from local misalignment: a node sits at a community interface when its membership composition differs meaningfully from that of its neighbors. A metric suited for boundary identification must therefore emphasize neighborhood-level discrepancies rather than departures from graph-wide averages. This motivates the development of a local quantity that captures how distinct a node is relative to the community structure surrounding it, independently of global centering and degree heterogeneity.

To capture neighborhood–level misalignment explicitly, we use a node attribute $s(\cdot)$ that measures alignment with neighbors. Following the pluralistic–homophily intuition—similarity via shared memberships at the dyadic level—we define

$$s(v) = \Big| \bigcup_{i \in N(v)} \big(\mathcal{C}(v) \cap \mathcal{C}(i)\big) \Big|, \tag{3}$$

Interpretation. By construction, $s(v)$ counts how many of the communities to which v belongs are represented at least once among its neighbors. In other words, for each community $c \in \mathcal{C}(v)$, we increment $s(v)$ by one if there exists at least one neighbor $u \in N(v)$ such that $c \in \mathcal{C}(u)$. No filtering or weighting is applied: $s(v)$ simply measures which portions of v's community identity are present in its immediate neighborhood.

We then define a purely local dissimilarity,

$$\delta_v = \frac{1}{d_v} \sum_{i \in N(v)} \big| \, s(v) - s(i) \, \big|, \tag{4}$$

where d_v is the degree of v.

We link local dissimilarity to a network-level reference via

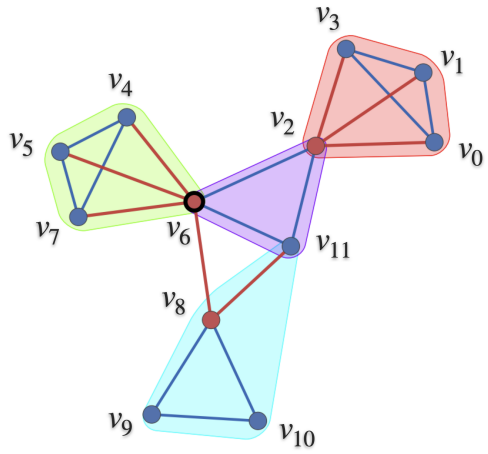
$$\lambda = \frac{h + \sum_u \delta_u}{N}, \tag{5}$$

where $N = |V|$ and h is the coefficient from Eq. 1 computed using the same attribute $s(\cdot)$ defined in Eq. 3. We then define the local score as

$$\tilde{h}_v = \lambda - \delta_v. \tag{6}$$

The local scores aggregate exactly to the network-level coefficient, i.e., $\sum_{v \in V} \tilde{h}_v = h$. A full proof is provided in the Supplementary Information (SI), Section S1.

This construction directly implements the boundary–detection intuition. The term $s(v)$ captures which portions of v's community identity are represented among its neighbors, and the quantity δ_v measures how different v is from the surrounding membership structure. The final score $\tilde{h}_v = \lambda - \delta_v$ normalizes these local discrepancies in a way that preserves the global coefficient ($\sum_v \tilde{h}_v = h$) while ensuring that the *ordering* of nodes is driven entirely by misalignment with their neighborhood. Nodes with strongly negative $\tilde{h}_v$ are precisely those whose memberships diverge from those of their neighbors, making them natural candidates for boundary–spanning roles in overlapping community structure.



(a)

Node	$\tilde{h}_v$	BridgeCC
v_0	0.012	0.0
v_1	0.012	0.0
v_2	-0.053	3.0
v_3	0.012	0.0
v_4	0.012	0.0
v_5	0.012	0.0
v_6	-0.069	**4.0**
v_7	0.012	0.0
v_8	-0.029	2.7
v_9	0.093	0.0
v_{10}	0.093	0.0
v_{11}	0.012	1.0

(b)

Figure 1. (a) Small overlapping network. Communities appear as colored polygons; nodes are colored by the sign of their local pluralistic homophily $\tilde{h}_v$ (blue: assortative, red: disassortative). The node with the largest BridgeCC is outlined in black. **(b)** Per-node values of neighborhood-based $\tilde{h}_v$ and the topology-based baseline BridgeCC.

Nature of the metric and topological comparator.

By construction, $\tilde{h}_v$ is a *membership–based* metric: it depends on community memberships in the neighborhood and is therefore *not* a purely topological score. To make this contrast explicit before turning to the illustrative example, we formally define the topology-based comparator BridgeCC as degree times one minus the local clustering coefficient:

$$\begin{aligned} \mathrm{BridgeCC}(v) &= d_v\big(1 - \mathrm{cc}(v)\big), \\ \mathrm{cc}(v) &= \frac{2T_v}{d_v(d_v - 1)}. \end{aligned} \tag{7}$$

where T_v is the number of triangles incident to v (with $\mathrm{cc}(v) = 0$ when $d_v < 2$). This formulation is motivated by the classic link between low transitivity and inter-community connectors [7] and by bridging-centrality formulations for bridge nodes [5,44]. This comparator emphasizes high-degree, low-clustering vertices—structural bridges connecting sparse regions—whereas $\tilde{h}_v$ targets membership- neighborhood misalignment.

Illustrative example.

Figure 1 contrasts $\tilde{h}_v$ with BridgeCC. Two interface nodes, v_2 and v_8, have negative $\tilde{h}_v$ because their incident edges connect neighborhoods with misaligned community profiles, whereas node v_6 maximizes BridgeCC (see Eq. (7)). This baseline prioritizes high–degree, low–clustering vertices, whereas $\tilde{h}_v$ highlights membership misalignment at community interfaces.

Importantly, in this example the strongest boundary position occurs at node v_6 for both metrics, but for opposite reasons: $\tilde{h}_v$ attains its *minimum* (more negative indicates stronger membership misalignment), whereas BridgeCC attains its *maximum*. Thus, the extrema of both metrics coincide at v_6, even though the two mechanisms they capture are conceptually orthogonal and need not align in general.

As this example illustrates, large negative $\tilde{h}_v$ values indicate membership disassortativity at the neighborhood level, characteristic of boundary positions where the community memberships of v differ from its neighbors.

Algorithm 1: Computation of Local Pluralistic Homophily Scores $\tilde{h}_v$

Input: Undirected graph $G = (V, E)$ with adjacency lists $N(v)$; community sets $\mathcal{C}(v)$ for all $v \in V$

Output: Local pluralistic homophily scores $\tilde{h}_v$ for all $v \in V$ and an optional ranking by ascending $\tilde{h}_v$

Step 1: Neighborhood-based membership alignment $s(v)$ **(Eq. 3)**;
foreach $v \in V$ **do**
  $S_{\text{neigh}}(v) \leftarrow \emptyset$;
  foreach $u \in N(v)$ **do**
    $S_{\text{neigh}}(v) \leftarrow S_{\text{neigh}}(v) \cup (\mathcal{C}(v) \cap \mathcal{C}(u))$;
  $s(v) \leftarrow |S_{\text{neigh}}(v)|$;

Step 2: Network-level pluralistic homophily h **(Eq. 1)**;
Compute the global pluralistic homophily coefficient h using Eq. 1 with $s(\cdot)$;

Step 3: Local dissimilarity δ_v **(Eq. 4)**;
foreach $v \in V$ **do**
  $d_v \leftarrow |N(v)|$;
  if $d_v = 0$ **then**
    $\delta_v \leftarrow 0$;
  else
    $\delta_v \leftarrow \frac{1}{d_v} \sum_{u \in N(v)} |s(v) - s(u)|$;

Step 4: Scaling factor λ **(Eq. 5)**;
$S_\delta \leftarrow \sum_{u \in V} \delta_u$;
$\lambda \leftarrow (h + S_\delta)/|V|$;

Step 5: Local pluralistic homophily scores $\tilde{h}_v$ **(Eq. 6)**;
foreach $v \in V$ **do**
  $\tilde{h}_v \leftarrow \lambda - \delta_v$;

Optional: ranking;
Sort all nodes by ascending $\tilde{h}_v$ (most negative first);

Algorithmic formulation.

Having illustrated the conceptual basis of $\tilde{h}_v$, we now detail its computation. Algorithm 1 provides a step-by-step procedure corresponding to Eqs. 3–6.

Computational complexity.

Computing δ_v for all nodes reduces to summing absolute differences of $s(\cdot)$ over incident edges, which is $O(M)$ once $s(\cdot)$ is precomputed. The attribute $s(v)$ (Eq. 3) requires traversing adjacency lists and community memberships, yielding overall complexity $O(M \cdot \bar{s})$, where $\bar{s}$ is the average membership count. For sparse networks with modest overlap ($\bar{s} < 10$ in our datasets), this scales essentially linearly in network size. A detailed analysis, including worst-case bounds and comparison with topology-based baselines, is provided in SI, Section S2. Empirical timing measurements on the eight real-world networks analyzed in this study (see Section S7) confirm the expected scaling behavior, with execution completing in under seven seconds for all networks except the largest-scale dataset.

Metric benchmarking under overlap.

We benchmark the unnormalized local contribution h_v (Eq. 2) against the normalized metric $\tilde{h}_v$ (Eq. 6) on empirical networks and on LFR benchmarks with controlled overlap [45]. Divergence increases with the prevalence of overlap, measured as the fraction of overlapping nodes: the Spearman rank correlation $\rho(h_v, \tilde{h}_v)$ declines markedly, indicating

that normalization induces a nontrivial reordering under high membership redundancy. As overlap grows, the proportion of disassortative nodes ($\tilde{h}_v < 0$) shrinks while the assortative region expands, consistent with $\tilde{h}_v$ trading recall for precision—fewer negative nodes overall, but more selective for boundary interfaces. For conceptual completeness, SI Section S3 documents this h_v–$\tilde{h}_v$ comparison with detailed analysis and visualizations; we do not revisit it thereafter, focusing instead on experimental design and empirical evaluation.

3.2. Datasets

We use eight real–world networks with overlapping communities. Four exhibit *interaction–based* communities—memberships emerging from sustained social or collaborative interactions: DBLP, LiveJournal (LJ), YouTube, and Twitch. Four exhibit *categorical* communities—memberships derived from metadata labels or platform taxonomies: Amazon, Deezer, StackOverflow (SO), and GitHub. For each dataset, we specify nodes, edges, and communities.

Interaction–based communities.

- **DBLP.** Nodes are authors; undirected edges denote co–authorship. Communities correspond to publication venues (conferences or journals); overlap arises when authors publish across venues.
- **LiveJournal (LJ).** Nodes are users; undirected edges represent mutual friendships. Communities are user–created interest groups; overlap arises when users join multiple groups.
- **YouTube.** Nodes are users; undirected edges represent friendship ties. Communities are subscription–based user groups; overlap arises when users subscribe to multiple groups.
- **Twitch.** Nodes are user accounts; undirected edges reflect interaction affinity (e.g., co–participation in chats, co–viewing, or follower ties). Communities are induced from streaming–channel or game–category participation; users often engage across multiple channels/categories, yielding overlap.

Categorical communities.

- **Amazon.** Nodes are products; undirected edges indicate co–purchase relations. Communities follow the platform product–category taxonomy; products may belong to multiple categories, producing overlap.
- **Deezer.** Nodes are users; undirected edges represent mutual friendships. Communities are induced from music–genre labels; users frequently belong to multiple genres, producing overlap.
- **StackOverflow (SO).** Nodes are users; undirected edges connect users who co–participate in the same thread via answers or comments. Communities are induced from question tags; users inherit tag memberships from their interactions with tagged posts.
- **GitHub.** Nodes are user accounts; undirected edges connect users who co–participate in the same repositories/issues. Communities are induced from repository metadata (topics, languages); users typically engage across multiple categories, producing overlap.

Community assignments are obtained from publicly available sources. For DBLP, LJ, and YouTube, we use the original group memberships from the source and build community sets as connected components within each provided group, following the SNAP repository [46]. For SO, Deezer, GitHub, Amazon, and Twitch, memberships are derived from platform labels or participation logs (e.g., tags, genres, topics, categories, channels); in all cases we extract connected components per label to form communities. SO

is derived from public StackExchange dumps [47]. All networks are treated as undirected and unweighted, and we retain nodes with multiple memberships to preserve overlap.

3.3. Experimental Setup and Evaluation Metrics

We evaluate whether the local pluralistic homophily score $\tilde{h}_v$ identifies functionally selective boundary nodes through a mechanism orthogonal to topology, and under which network conditions this mechanism yields advantages. The setup has three parts: data standardization, node ranking, and two complementary evaluations—a mechanistic test on inter–community coupling and a functional test on interface selectivity.

Data standardization.

We load undirected, unweighted graphs with overlapping community assignments, where communities are taken as connected components within each group, keeping components of size ≥ 3 to preserve overlap while avoiding trivial communities. The resulting preprocessed edge lists and community assignments used in this study are publicly archived (see Data Availability Statement) for long-term reproducibility.

Overlap descriptors.

Let $\mathcal{C}(v)$ denote the membership set of node v and $m(v) = |\mathcal{C}(v)|$ its membership count. We characterize overlap using two complementary descriptors: the mean memberships per node $\langle m \rangle = \frac{1}{|V|} \sum_v m(v)$ captures the average intensity of overlap, while the overlap prevalence

$$\psi = \frac{|\{\, v \in V : \; m(v) \geq 2 \,\}|}{|V|} \tag{8}$$

captures its spatial extent (the fraction of nodes belonging to two or more communities). Together, these metrics distinguish networks where overlap is rare but intense from those where it is widespread but modest. Both descriptors are reported in Table 1.

Node ranking by $\tilde{h}_v$.

For each node v, we compute $\tilde{h}_v$ as defined in Section 3. We then fix a deterministic order by ascending $\tilde{h}_v$ (most negative first) and keep this order fixed for all removal budgets $p \in [0, 1]$ used downstream.

Mechanistic test: inter–community coupling under targeted removal.

Let G_p be the graph obtained after removing the first $k(p) = \lceil p|V| \rceil$ nodes in the fixed order, and let $\mathcal{C}(G_p)$ denote its communities. We summarize inter–community connectivity via a community–level graph whose weighted edges count links in G_p that span community boundaries. For distinct $c_a, c_b \in \mathcal{C}(G_p)$, let $W_{c_a c_b}(p)$ be the corresponding cross–community weight. The aggregate inter–community weight and its normalized trajectory are

$$W_{\text{inter}}(G_p) = \tfrac{1}{2} \sum_{c_a \neq c_b} W_{c_a c_b}(p), \qquad \widehat{W}_{\text{inter}}(p) = \frac{W_{\text{inter}}(G_p)}{W_{\text{inter}}(G_0)},$$

with $G_0 \equiv G$ so that $\widehat{W}_{\text{inter}}(0) = 1$.

Weights are symmetric ($W_{c_a c_b} = W_{c_b c_a}$) and we exclude self–loops; the $\frac{1}{2}$ factor avoids double counting. The edge-community incidence matrix and community graph construction are formalized in SI, Section S5.

We compare the targeted trajectory against a random baseline constructed from K independent uniform permutations of the node order (pointwise mean μ_{rand} and $[P_5, P_{95}]$

band). Two summaries are reported: the separation budget p_{sep} (earliest p with three consecutive points below P_5) and the area difference:

$$\Delta\text{AUC} = \int_0^1 \left(\widehat{W}_{\text{inter}}^{\text{target}}(p) - \mu_{\text{rand}}(p)\right) dp,$$

where negative values indicate faster depletion than random.

As a connectivity control, we measure the giant–component fraction $\rho_{\text{GCC}}(p)$ to ensure that reductions in $\widehat{W}_{\text{inter}}(p)$ do not result from premature fragmentation.

Functional test: selectivity at community interfaces.

We compare the selections induced by $\tilde{h}_v$ against two complementary baselines: the topology-based BridgeCC (Eq. 7, Section 3.1) and a community-aware baseline denoted mDC. While BridgeCC targets structural sparsity through degree and clustering, mDC operates directly on overlapping community structure and ranks nodes according to the density of their intra-community connections. Intuitively, mDC assigns higher scores to nodes whose incident edges concentrate within the communities to which they belong, and lower scores to nodes occupying more weakly embedded positions. Because mDC relies solely on community information—without capturing membership–neighborhood misalignment—it serves as a natural community-aware reference for evaluating whether the pluralistic-homophily mechanism provides additional interface selectivity.

We label an edge (u, v) as inter-community when the Jaccard similarity between endpoint membership sets satisfies $J(\mathcal{C}(u), \mathcal{C}(v)) < \tau$, with τ=0.2 (rationale below). For a selection S at a matched budget, we compute

$$\text{IER}_{\text{node-mean}} = \frac{1}{|S|} \sum_{v \in S} \frac{\deg_{\text{inter}}(v)}{\max\{1, \deg(v)\}}, \qquad \text{IER}_{\text{stub}} = \frac{\sum_{v \in S} \deg_{\text{inter}}(v)}{\sum_{v \in S} \deg(v)}.$$

We report both aggregations because they capture complementary aspects of interface selectivity. $\text{IER}_{\text{node-mean}}$ quantifies typical boundary purity at the node level, measuring whether selected nodes are themselves boundary-focused. IER_{stub} quantifies total boundary capacity, measuring the aggregate cross-group edge volume controlled by selected nodes. We compare the selections induced by $\tilde{h}_v$, the topology-based baseline BridgeCC, and the community-aware baseline mDC at matched budgets. Node-mean IER is reported for all three strategies, whereas stub-weighted capacity (IER_{stub}) is reported only for $\tilde{h}_v$ and BridgeCC, which are explicitly designed to reflect global inter–community flow. Uncertainty is quantified with node-bootstrap confidence intervals. Degree-stratified permutation tests provide two-sided p values.

Rationale for τ=0.2. We set τ=0.2 to identify low–overlap interfaces where boundary definitions remain interpretable. An edge is considered inter–community when $J(\mathcal{C}(u), \mathcal{C}(v)) < 0.2$, corresponding to at most 20% shared memberships between endpoints. This excludes (i) trivially disjoint cases ($J \approx 0$), which align with structural–hole interpretations, and (ii) high–overlap cases ($J > 0.3$), where boundary signals become diluted. This choice follows established practice in the use of low–overlap thresholds for identifying cross–community edges [7,48].

To examine robustness with respect to this design choice, we recomputed the node–mean inter–community edge ratio (IER) for $\tilde{h}_v$ under matched–budget removal for $\tau \in \{0.1, 0.2, 0.3\}$. Detailed values are reported in the SI (Section S8). Across all datasets, $\tilde{h}_v$ exhibits monotonic and stable behavior over this range of thresholds, supporting τ=0.2 as a representative and balanced operating point for evaluating interface selectivity.

Implementation and availability.

All steps—standardization, scoring, and evaluation—are scripted end to end. Parameter choices (budgets, number of permutations, bootstrap replicates, and τ=0.2) and random seeds are documented in a public GitHub repository (see Data Availability Statement).

4. Results

We report: (i) dataset descriptors; (ii) ranking divergence between $\tilde{h}_v$ and topology-based metrics; (iii) budgeted evaluation of inter–community coupling depletion under targeted removal; and (iv) functional interface selectivity comparing $\tilde{h}_v$ against both a topology-based baseline (BridgeCC) and a community-aware baseline (mDC).

4.1. Network and Community Descriptors

Table 1 reports structural and community-level metrics for the eight datasets. In addition to standard topology descriptors (nodes, edges, degree) and community counts, we report overlap-specific metrics: $\langle m \rangle$ (mean memberships per node) and ψ (overlap prevalence, Eq. 8). Network-level pluralistic homophily $\tilde{h}$[1] ranges from -0.076 (SO) to 0.489 (Amazon).

Table 1. Network and community descriptors. N: nodes; E: edges; $\langle d \rangle$: mean degree; $|C|$: communities; $\langle m \rangle$: mean memberships per node; ψ: overlap prevalence; $\tilde{h}$: network-level pluralistic homophily.

Dataset	N	E	$\langle d \rangle$	$\lvert C \rvert$	$\langle m \rangle$	ψ	$\tilde{h}$
YouTube	1,134,890	2,987,624	5.27	8,385	2.40	0.02	0.068
LJ	3,997,962	34,681,189	17.35	287,512	5.86	0.18	0.223
Twitch	9,498	153,138	32.25	9,498	76.82	0.22	0.005
DBLP	317,080	1,049,866	6.62	13,477	2.76	0.35	0.217
Deezer	54,573	498,202	18.26	1,711	5.22	0.76	0.049
Amazon	334,863	925,872	5.53	75,149	7.16	0.90	0.489
SO	2,330,420	12,802,064	10.99	161,462	9.95	0.91	-0.076
GitHub	37,700	289,003	15.33	3,882	18.31	1.00	-0.072

Datasets exhibit substantial heterogeneity across structural and overlap dimensions. Network size spans over two orders of magnitude ($N \in [9.5\text{k}, 4.0\text{M}]$), with corresponding variation in edge counts ($E \in [153\text{k}, 34.7\text{M}]$) and community counts ($|C| \in [1.7\text{k}, 287\text{k}]$). Mean degree ranges from 5.27 (YouTube) to 32.25 (Twitch), reflecting diverse connectivity patterns.

Overlap structure varies markedly across datasets: minimal in YouTube (ψ=0.02), low–moderate in LJ (ψ=0.176) and Twitch (ψ=0.22), moderate in DBLP (ψ=0.35), high in Deezer (ψ=0.76), very high in Amazon (ψ=0.90) and SO (ψ=0.91), and maximal in GitHub (ψ=1.00). Ground-truth datasets span minimal–very-high overlap (YouTube, LJ, DBLP, Amazon), whereas metadata-derived networks concentrate at high-to-maximal levels (Deezer, SO, GitHub), with one low–moderate case (Twitch). Network-level homophily is positive in six cases ($\tilde{h} \in [0.005, 0.489]$), indicating assortative membership patterns, while SO and GitHub exhibit mildly disassortative behavior ($\tilde{h} \in [-0.076, -0.072]$).

Together, these descriptors encompass overlap regimes ranging from near-absent to maximal ($\psi \approx 0.02$ to 1.0), providing a broad testbed for subsequent comparisons between membership-based and topology-based rankings.

[1] The network-level coefficient $\tilde{h}$ corresponds to the network-wide aggregation of the local metric $\tilde{h}_v$, a relationship ensured by the scaling factor λ (Eq. 5). This equivalence provides a consistent baseline for comparing pluralistic homophily at the network level.

Table 2. Spearman rank correlations between $\tilde{h}_v$ and topology-based metrics. Negative values indicate that low-$\tilde{h}_v$ nodes occupy positions distinct from high-centrality hubs and structural-bridge nodes; positive values indicate alignment.

Dataset	BridgeCC	Degree	Eigenvector
DBLP	−0.213	−0.114	−0.687
LJ	−0.394	−0.347	−0.694
YouTube	−0.334	−0.377	−0.578
SO	−0.080	−0.070	−0.692
Deezer	−0.302	−0.304	−0.260
GitHub	0.407	0.387	0.100
Amazon	−0.137	−0.057	−0.300
Twitch	−0.397	−0.435	−0.533
Mean	−0.181	−0.165	−0.455

4.2. Ranking Divergence from Topology-Based Baselines

We quantify the extent to which $\tilde{h}_v$ and topology-based metrics identify distinct node populations via Spearman rank correlations. Table 2 reports correlations between $\tilde{h}_v$ and three topology-based references: degree, eigenvector, and BridgeCC (our main baseline).

Associations are predominantly negative in seven of eight datasets, with mean correlations $\rho_s = -0.181$ (BridgeCC), -0.165 (degree), and -0.455 (eigenvector). Negative coefficients indicate that low-$\tilde{h}_v$ nodes systematically occupy positions distinct from high–centrality hubs and structural–bridge nodes. GitHub presents a reversed pattern, with positive correlations $\rho_s \in [0.100, 0.407]$.

Synthetic Benchmarks: LFR Rank Divergence under Overlap

To test whether rank divergence generalizes beyond empirical data, we replicate the ranking analysis on LFR benchmarks with tunable overlap. As overlap increases, Spearman correlations between $\tilde{h}_v$ and topology-based baselines (degree, eigenvector, BridgeCC) remain weak and tend toward zero, indicating that membership-based rankings diverge from structural heuristics as communities overlap increases. Details are provided in SI, Section S4 (Table S2).

4.3. Inter-Community Coupling Depletion

We evaluate budgeted node removal using a fixed ranking by ascending local pluralistic homophily $\tilde{h}_v$ (lowest $\tilde{h}_v$ first). For each budget $p \in [0, 1]$, we remove the pN lowest-ranked nodes and recompute the normalized inter–community coupling $\widehat{W}_{\text{inter}}(p)$ on the community–level graph. As a baseline, we report K-fold random removal (K=100), summarized by the pointwise mean and the 5th–95th percentile band.

Figure 2 shows $\widehat{W}_{\text{inter}}(p)$ across eight datasets. Networks with interaction–based communities (LJ, YouTube, DBLP, Twitch) exhibit early separation from the random baseline, with user–created networks separating at p_{sep}=0.06 and showing substantial aggregate depletion ($\Delta\text{AUC} \in [-0.215, -0.135]$; Table 3). LJ shows the steepest initial drop, DBLP maintains a stable gap, and YouTube depletes early and then plateaus. By contrast, networks with categorical communities (Amazon, Deezer, SO, GitHub) display attenuated or delayed effects, with Amazon presenting a more gradual depletion profile.

As a connectivity control, we measure the giant–component fraction $\rho_{\text{GCC}}(p)$ to ensure that reductions in $\widehat{W}_{\text{inter}}(p)$ do not arise from premature network collapse. Across all eight datasets, fractions remain high at early budgets: $\rho_{\text{GCC}}(0.05) \in [0.87, 0.95]$, declining to 0.76–0.90 at p=0.10 (Table 3). YouTube exhibits the steepest early decline (to 0.76 at p=0.10) but remains well above fragmentation thresholds.

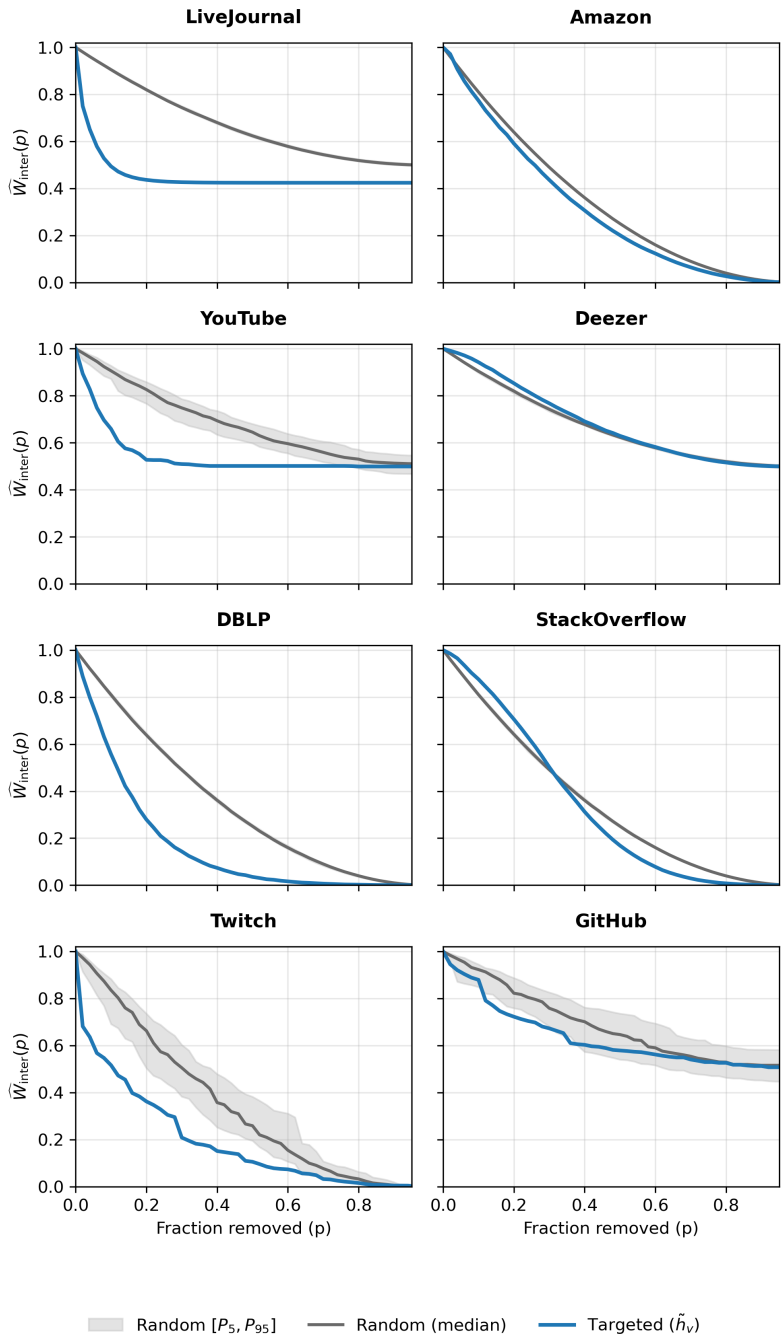


Figure 2. Inter–community coupling depletion under targeted removal across eight networks. Left column: interaction-based communities (LiveJournal, YouTube, DBLP, Twitch). Right column: categorical communities (Amazon, Deezer, StackOverflow, GitHub). Normalized inter-community weight $\widehat{W}_{\text{inter}}(p)$ versus fraction removed p. Blue: targeted removal by ascending $\tilde{h}_v$; dark gray: random-removal median; light gray band: $[P_5, P_{95}]$ (K=100 folds). All panels are normalized at p=0 and share the same vertical scale.

Networks where communities arise from sustained interactions show consistent early separation and strong aggregate depletion. YouTube (ψ=0.02, ΔAUC=$-$0.135), LiveJournal (ψ=0.18, ΔAUC=$-$0.215), and DBLP (ψ=0.35, ΔAUC=$-$0.159) separate at p_{sep}=0.06, with LJ exhibiting the steepest initial drop, DBLP maintaining a stable gap, and YouTube depleting early before plateauing. The social platform Twitch, where game communities emerge from viewing interactions, follows the same pattern with early separation and substantial depletion (p_{sep}=0.06, ΔAUC=$-$0.151).

Table 3. Coupling depletion and connectivity control under targeted removal. p_{sep}: earliest budget with $\geq$3 consecutive points below the random $[P_5, P_{95}]$ band (—: criterion not met). ΔAUC: integrated gap (targeted minus random) over $p \in [0,1]$; negative values indicate faster depletion. $\rho_{\mathrm{GCC}}(0.05)$ and $\rho_{\mathrm{GCC}}(0.10)$: giant–component fractions at early budgets. Rows are sorted by ΔAUC (more negative first).

Dataset	p_{sep}	**ΔAUC**	$\rho_{\mathrm{GCC}}(0.05)$	$\rho_{\mathrm{GCC}}(0.10)$
LJ	0.06	−0.215	0.93	0.86
DBLP	0.06	−0.159	0.93	0.85
Twitch	0.06	−0.151	0.95	0.90
YouTube	0.06	−0.135	0.87	0.76
GitHub	0.16	−0.051	0.94	0.88
Amazon	0.08	−0.032	0.94	0.88
SO	0.38	−0.016	0.94	0.88
Deezer	—	+0.012	0.95	0.90

By contrast, informative or category–driven systems display attenuated or mixed effects. Amazon (curated product categories) shows a moderate aggregate effect (p_{sep}=0.08, ΔAUC≈−0.032); GitHub separates later with moderate depletion (p_{sep}=0.16, ΔAUC=−0.051); SO separates late and weakly (p_{sep}=0.38, ΔAUC=−0.016); and Deezer does not meet the separation criterion, with slightly weaker aggregate performance (ΔAUC=+0.012).

4.4. Functional Selectivity at Community Interfaces

We assess interface selectivity using the inter–community edge ratio (IER) at the Jaccard threshold τ=0.2, as defined in §3. We focus on the per–node mean aggregation ($\mathrm{IER}_{\text{node-mean}}$), and compare $\tilde{h}_v$ against both a topology-based baseline (BridgeCC) and a community-aware baseline (mDC). Figure 3 summarizes the node-mean differences, while Table 4 reports absolute values for all strategies and the corresponding degree-weighted results ($\mathrm{IER}_{\mathrm{stub}}$).

Relative to BridgeCC (Fig. 3A), $\tilde{h}_v$ improves $\mathrm{IER}_{\text{node-mean}}$ in seven of eight networks at τ=0.2: large gains in DBLP (+25.6 pp), GitHub (+25.3 pp), and Amazon (+24.0 pp); moderate in Deezer (+5.3 pp) and Twitch (+4.3 pp); and small in YouTube (+1.1 pp) and SO (+1.1 pp). LiveJournal is the only case where BridgeCC yields higher per-node purity (−8.7 pp). Degree-weighted results ($\mathrm{IER}_{\mathrm{stub}}$; Table 4) favor $\tilde{h}_v$ in all eight networks—including LiveJournal (+15.6 pp), Amazon (+21.8 pp), and Twitch (+10.5 pp)—indicating greater cross-group capacity even when per-node purity favors the topological baseline.

When compared to the community-aware baseline mDC (Fig. 3B), $\tilde{h}_v$ remains competitive or superior in most networks. Node-mean advantages are large in DBLP (+25.9 pp), GitHub (+25.7 pp), and Amazon (+18.3 pp), and moderate in Deezer (+4.8 pp), Twitch (+4.3 pp), and SO (+3.1 pp). In YouTube and LiveJournal, mDC attains slightly higher node-mean purity (−0.4 pp and −0.3 pp, respectively), but these effects are small in magnitude compared to the gains observed in high-overlap, interaction-driven systems. Overall, the strongest advantages of $\tilde{h}_v$ over mDC are observed precisely in the networks where inter-community coupling depletion is most pronounced (DBLP, GitHub, Amazon), indicating that pluralistic homophily not only disrupts community interfaces more efficiently (Fig. 2) but also yields sharper interface composition.

As an external coherence check across datasets, we correlated interface selectivity with depletion strength ($D \equiv -\Delta\mathrm{AUC}$). At τ=0.2, associations are modest (Pearson $r \approx -0.30$ for node–mean, $r \approx +0.30$ for stub-weighted), reflecting the purity–capacity trade–off. Associations remain comparable at $\tau \in \{0.1, 0.3\}$, confirming that τ=0.2 lies within a stability plateau.

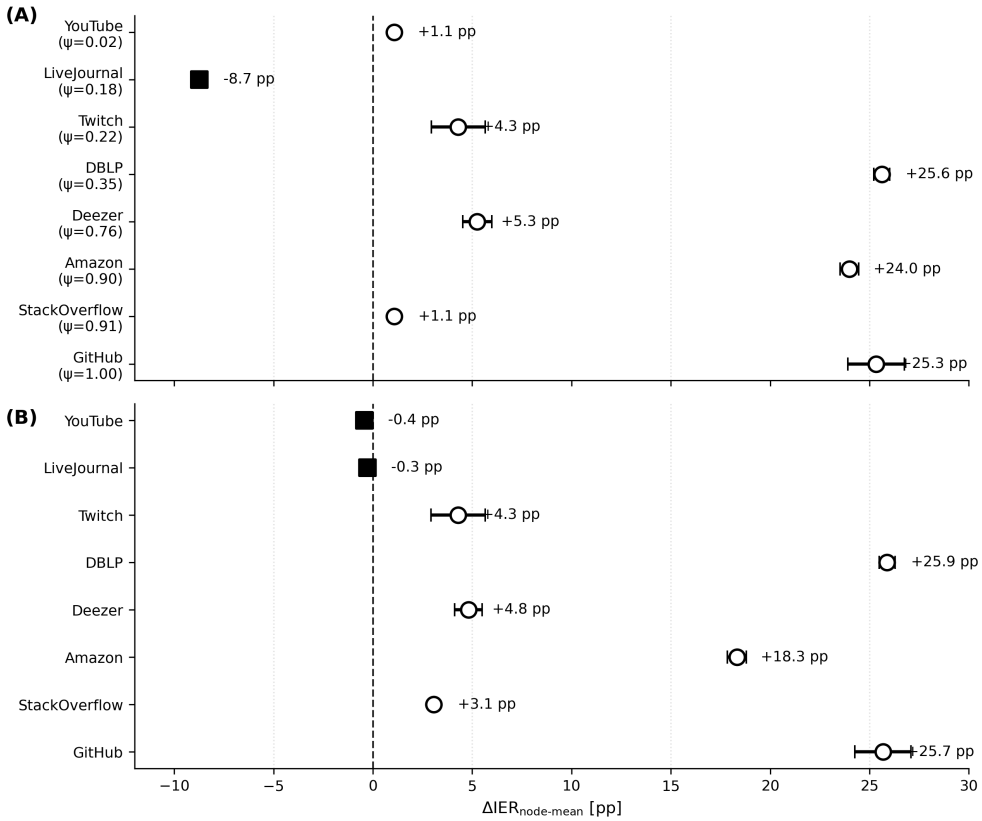


Figure 3. Inter–community edge ratio advantage at τ=0.2 (node–mean aggregation). Panel (A): $\Delta\mathrm{IER}_{\text{node-mean}}$ for $\tilde{h}_v$ versus the topology-based baseline BridgeCC; Panel (B): $\Delta\mathrm{IER}_{\text{node-mean}}$ for $\tilde{h}_v$ versus the community-aware baseline mDC. Points show differences in percentage points; circles indicate $\tilde{h}_v$ advantage and squares indicate baseline advantage. Horizontal lines denote 95% bootstrap confidence intervals, and the vertical dashed line marks the null. Datasets are ordered by absolute effect magnitude and annotated with overlap prevalence ψ. Value labels are shown only when $|\Delta\mathrm{IER}| > 5$ pp; stub-weighted results are reported in Table 4.

Table 4. Inter–community edge composition at τ=0.2. $\mathrm{IER}_{\text{node-mean}}$ (per–node purity) for $\tilde{h}_v$, BridgeCC, and mDC, and $\mathrm{IER}_{\text{stub}}$ (total capacity) for $\tilde{h}_v$ and BridgeCC. Δ_{B} denotes the node–mean advantage of $\tilde{h}_v$ over the topology-based baseline (BridgeCC), and Δ_{C} denotes its advantage over the community-aware baseline (mDC).

	$\mathrm{IER}_{\text{node-mean}}$					$\mathrm{IER}_{\text{stub}}$		
Dataset	$\tilde{h}_v$	**BridgeCC**	**mDC**	Δ_{B}	Δ_{C}	$\tilde{h}_v$	**BridgeCC**	Δ_{B}
DBLP	0.766	0.510	0.507	+0.256	+0.259	0.778	0.580	+0.198
LiveJournal	0.457	0.544	0.460	−0.087	−0.003	0.747	0.591	+0.156
YouTube	0.056	0.046	0.061	+0.011	−0.004	0.166	0.151	+0.015
StackOverflow	0.963	0.952	0.932	+0.011	+0.031	0.976	0.973	+0.003
Deezer	0.497	0.445	0.449	+0.053	+0.048	0.467	0.447	+0.019
GitHub	0.720	0.467	0.463	+0.253	+0.257	0.684	0.501	+0.182
Amazon	0.408	0.168	0.225	+0.240	+0.183	0.395	0.177	+0.218
Twitch	0.114	0.071	0.071	+0.043	+0.043	0.177	0.071	+0.105

Sensitivity to τ.

Replicating the analysis at $\tau \in \{0.1, 0.3\}$ confirms that qualitative patterns persist, while revealing distinct boundary characteristics. Amazon's $\Delta\text{IER}_{\text{node-mean}}$ increases monotonically with τ (+18.9 pp $\rightarrow$ +24.7 pp), consistent with sharply defined curated categories; SO and Deezer invert from positive at τ=0.1 to negative at τ=0.3, indicating diffuse tag/genre boundaries; DBLP remains positive but attenuates (+37.7 pp $\rightarrow$ +11.7 pp) as stricter thresholds exclude legitimate overlap interfaces; and LJ remains negative at the node-mean while staying positive under stub weighting across all τ, confirming a purity–capacity trade–off rather than a threshold artifact.

Taken together, per-node selectivity improves in seven of eight networks and stub-weighted selectivity in all eight (including LiveJournal) when comparing $\tilde{h}_v$ to the topology-based baseline, and $\tilde{h}_v$ remains competitive or superior to the community-aware baseline in six of eight networks. This provides a concise empirical summary of the comparison against both topology-based and community-aware baselines.

5. Discussion

Overall, $\tilde{h}_v$ improves interface selectivity in seven of eight networks at the per–node level and in all eight under stub–weighted aggregation, indicating effective boundary targeting without implying universal dominance over topology–based or community–aware baselines. Across all datasets, removing nodes in ascending $\tilde{h}_v$ depletes the normalized inter–community weight $\widehat{W}_{\text{inter}}(p)$ earlier and more strongly than random, while the giant–component fraction $\rho_{\text{GCC}}(p)$ remains high at early budgets. This joint pattern indicates that targeted removal concentrates on cross–community links without prematurely fragmenting the graph—consistent with selecting boundary–spanning nodes. Rank correlations with topology–based metrics are predominantly negative (Table 2), reinforcing that $\tilde{h}_v$ identifies boundary positions that differ systematically from structural hubs.

Selectivity analyses further clarify these distinctions. At matched budgets, the inter–community edge ratio (IER) at a low–overlap threshold (τ=0.2) favors $\tilde{h}_v$ over both the topology–based comparator (BridgeCC) and the community–aware baseline (mDC) in most datasets. Per–node purity ($\text{IER}_{\text{node-mean}}$) improves in seven of eight networks, while aggregate boundary capacity (IER_{stub}) increases in all eight. Cases where node–mean purity reverses (e.g., LiveJournal) still show gains in aggregate capacity, revealing a purity–capacity trade–off: BridgeCC isolates a few high–purity nodes, while $\tilde{h}_v$ distributes cross–community load across many membership–misaligned neighborhoods. In LiveJournal (ψ=0.18), this mechanism produces strong boundary–capacity depletion (ΔAUC=-0.215) even though per–node purity favors BridgeCC. Conversely, in networks with minimal overlap (YouTube, ψ=0.02), all methods yield similar node–mean selectivity because few edges satisfy the inter–community criterion at τ=0.2, reducing discriminability—consistent with the broader observation that community-awareness alone is insufficient when interfaces are extremely sparse.

Separation thresholds further reveal where boundary capacity concentrates (Table 3). Networks with interaction–based memberships (LJ, YouTube, DBLP, Twitch) show early separation and strong depletion (p_{sep}=0.06, $\Delta\text{AUC} \in [-0.215, -0.135]$), indicating that membership–misaligned nodes play a central role in cross–group connectivity. Categorical networks exhibit more heterogeneous behavior: sharp boundaries yield strong selectivity (Amazon), diffuse taxonomies yield weaker effects (SO, Deezer), and intermediate structure yields moderate performance (GitHub). These patterns suggest that $\tilde{h}_v$ is most effective where membership assignments reflect cohesive behavioral or semantic groupings, while topology–based baselines may excel when boundary roles concentrate in a small number of structurally distinctive vertices.

In practice, $\tilde{h}_v$ provides advantages in settings with coherent, interaction–driven communities (e.g., DBLP, LJ in capacity, Twitch) and in categorical systems with sharp interfaces (Amazon), while BridgeCC and related structural methods remain competitive across a broad range of overlap regimes. The community–aware baseline mDC [10] performs well when intra–community density is the dominant signal, but lacks the neighborhood–alignment mechanism necessary to capture membership–profile misalignment at local interfaces. The complementarity among these mechanisms explains why $\widehat{W}_{\text{inter}}(p)$ depletes without immediate fragmentation: structural hubs generally remain intact while cross–community connectors are removed. This is relevant for applications where reducing cross–group exposure must be balanced against preserving within–group cohesion [33,36].

Table 5 situates our approach relative to established bridge–detection families.

Table 5. Positioning relative to established bridge–detection approaches. Our metric directly exploits overlapping memberships, is unsupervised, and runs in linear time.

Method class	Input	Complexity	Community-based	Overlap-based	Unsupervised
Betweenness [4]	Topology	$O(NM)$	No	No	Yes
BridgeCC [5]	Topology	$O(\sum d_v^2)$	No	No	Yes
DoN/mDC [10]	Topology+Communities	$O(M)$	Yes	Partial	Yes
GNN-based [31,32]	Topology+Features	Training	Yes	No	No
$\tilde{h}_v$ (this work)	Topology+Communities	$\mathbf{O(M)}$	Yes	Yes	Yes

Limitations.

Our method inherits the fundamental dependency on community assignments shared by all community–aware approaches. Because $\tilde{h}_v$ measures membership–neighborhood misalignment, its interpretation and performance depend on the quality of the underlying communities. In this study, we use both ground–truth memberships (e.g., publication venues in DBLP or user–created groups in LiveJournal) and proxy or metadata–derived assignments (e.g., tags, genres, product categories), and we observe consistent patterns across both types. Nevertheless, the quality of detected communities can vary substantially across algorithms, especially under high overlap, heterogeneous densities, or resolution–limit effects [45,48].

When using algorithmically detected communities, several sources of variability may affect $\tilde{h}_v$: communities may fragment (splitting cohesive groups), merge (combining structurally distinct clusters), or produce boundary artifacts due to resolution biases or thresholding [49]. Such distortions may attenuate or amplify membership–neighborhood misalignment, affecting the magnitude of $\tilde{h}_v$ while preserving its qualitative ranking of true boundary nodes. These issues similarly affect mDC and any method relying on community structure, though $\tilde{h}_v$ remains locally interpretable because it evaluates neighborhood alignment rather than global membership counts.

In practice, the dependency on community quality suggests two considerations. First, $\tilde{h}_v$ should be applied in settings where memberships reflect semantically or behaviorally cohesive groups, whether human–defined or algorithmically detected with robust overlap handling. Second, when community detection is required, overlap–aware algorithms (e.g., link partitioning [48] or LFR–compatible methods [45]) provide more reliable boundary structure than disjoint or resolution–limited approaches. Although community quality conditions the strength of observed effects, the joint evidence from ground–truth and metadata–derived networks in this study shows that the pluralistic–homophily mechanism remains stable across heterogeneous membership sources. Future work could systematically quantify this dependency by comparing $\tilde{h}_v$ performance across multiple detection algorithms on controlled benchmarks, though such validation is beyond the scope of this study.

Future work.

The node–level formulation naturally extends to edge– and community–level analogues ($\tilde{h}_e$, $\tilde{h}_C$), preserving normalization and additivity with the global coefficient. These variants could support multi–scale boundary analysis, capturing interface alignment across edges and cohesion within overlapping groups. Embedding $\tilde{h}_v$ into diffusion models (e.g., SIR, SEIR) would quantify boundary–sensitive outcomes such as cross–group transmission and exposure inequality, comparing targeted interventions against topology–based controls. Additionally, extending the method to weighted and directed networks or to settings with algorithmically detected communities will clarify robustness and transferability across diverse systems.

Supplementary Materials: Supplementary Information is included at the end of this article

Author Contributions: Conceptualization, F.B., C.R.-O., A.A. and A.F.; Methodology, F.B., C.R.-O., and A.A.; Software, F.B.; Validation, F.B., C.R.-O. and A.A.; Formal analysis, F.B., C.R.-O. and A.A.; Data curation, F.B.; Visualization, F.B.; Supervision, A.F.; Project administration, F.B.; Writing—original draft preparation, F.B.; Writing—review and editing, F.B., C.R.-O., A.A. and A.F. All authors have read and agreed to the published version of the manuscript.

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Supplementary Information

S1. Local pluralistic homophily: formal derivations

This section provides the detailed mathematical derivations and proofs supporting the local pluralistic homophily metric $\tilde{h}_v$ and its properties.

Derivation, equivalent forms, and global–local consistency of the local metric.

We define

$$\tilde{h}_v = \lambda - \delta_v, \tag{9}$$

with

$$\lambda = \frac{h + \sum_u \delta_u}{N}. \tag{10}$$

Here, h is the network-level pluralistic homophily coefficient, δ_u is the local dissimilarity of node u, and N is the number of nodes. **Breaking the sum in λ.** Since

$$\sum_u \delta_u = \delta_v + \sum_{u \neq v} \delta_u,$$

substituting into Eq. (10) gives

$$\lambda = \frac{h + \delta_v + \sum_{u \neq v} \delta_u}{N}.$$

Expanded expression for $\tilde{h}_v$: Plugging into Eq. (9) yields

$$\begin{aligned}\tilde{h}_v &= \left(\frac{h + \delta_v + \sum_{u \neq v} \delta_u}{N}\right) - \delta_v = \frac{h + \sum_{u \neq v} \delta_u}{N} + \frac{\delta_v}{N} - \delta_v \\ &= \frac{h + \sum_{u \neq v} \delta_u}{N} - \delta_v \left(1 - \frac{1}{N}\right),\end{aligned}$$

so $\tilde{h}_v$ depends on h, $\sum_{u \neq v} \delta_u$, and δ_v scaled by N (with $\delta_v \geq 0$ measuring dissimilarity from neighbors).

Large-N asymptotics: For large N (so $1/N \approx 0$),

$$\tilde{h}_v \approx \frac{h + \sum_{u \neq v} \delta_u}{N} - \delta_v.$$

Global–local consistency ($\sum_v \tilde{h}_v = h$): Summing Eq. (9) over v,

$$\begin{aligned}\sum_v \tilde{h}_v &= \sum_v (\lambda - \delta_v) = N\lambda - \sum_v \delta_v \\ &= N \cdot \frac{h + \sum_u \delta_u}{N} - \sum_v \delta_v = (h + \sum_u \delta_u) - \sum_v \delta_v,\end{aligned}$$

and since $\sum_u \delta_u = \sum_v \delta_v$, it follows that $\sum_v \tilde{h}_v = h$.

Generalized assortativity for a real attribute: equivalence of formulations.

We verify the equivalence of two expressions for assortativity r for a continuous attribute $s : V \to \mathbb{R}$: Let $G = (V, E)$ be undirected with $|E| = M$ and $s : V \to \mathbb{R}$. Consider the two forms:

$$r = \frac{\sum_{x,y} s_x s_y (e_{xy} - q_x q_y)}{\sigma_q^2}, \tag{11}$$

$$r = \frac{\sum_{(i,j) \in E} (s(i) - \mu_q)(s(j) - \mu_q)}{\sum_{(i,j) \in E} (s(i) - \mu_q)^2}. \tag{12}$$

Definitions:

$$e_{xy} = \frac{|\{(i,j) \in E : s(i) = s_x,\ s(j) = s_y\}|}{M}, \quad q_x = \sum_y e_{xy}, \quad \mu_q = \sum_x s_x q_x = \frac{1}{2M} \sum_{(i,j)\in E} \big(s(i) + s(j)\big),$$

$$\sigma_q^2 = \sum_x (s_x - \mu_q)^2 q_x = \frac{1}{2M} \sum_{(i,j)\in E} \big[(s(i) - \mu_q)^2 + (s(j) - \mu_q)^2\big].$$

Numerator of Eq. (11):

$$\sum_{x,y} s_x s_y (e_{xy} - q_x q_y) = \sum_{x,y} s_x s_y e_{xy} - \sum_{x,y} s_x s_y q_x q_y = \frac{1}{M} \sum_{(i,j)\in E} s(i)s(j) - \mu_q^2.$$

Centering: Using

$$s(i)s(j) = (s(i) - \mu_q)(s(j) - \mu_q) + \mu_q(s(i) - \mu_q) + \mu_q(s(j) - \mu_q) + \mu_q^2,$$

and

$$\sum_{(i,j)\in E} (s(i) - \mu_q) = \sum_{(i,j)\in E} (s(j) - \mu_q) = 0,$$

we obtain

$$\sum_{(i,j)\in E} \big(s(i)s(j) - \mu_q^2\big) = \sum_{(i,j)\in E} (s(i) - \mu_q)(s(j) - \mu_q).$$

Back substitution:

$$\sum_{x,y} s_x s_y (e_{xy} - q_x q_y) = \frac{1}{M} \sum_{(i,j)\in E} (s(i) - \mu_q)(s(j) - \mu_q).$$

Equality of denominators: Since

$$\sigma_q^2 = \frac{1}{2M} \sum_{(i,j)\in E} \big[(s(i) - \mu_q)^2 + (s(j) - \mu_q)^2\big],$$

it follows that

$$M\sigma_q^2 = \frac{1}{2} \sum_{(i,j)\in E} \big[(s(i) - \mu_q)^2 + (s(j) - \mu_q)^2\big] = \sum_{(i,j)\in E} (s(i) - \mu_q)^2,$$

because each node appears equally often as i and as j across edges. Therefore,

$$r = \frac{\sum_{(i,j)\in E} (s(i) - \mu_q)(s(j) - \mu_q)}{M\sigma_q^2} = \frac{\sum_{(i,j)\in E} (s(i) - \mu_q)(s(j) - \mu_q)}{\sum_{(i,j)\in E)(s(i)-\mu_q)^2}},$$

which completes the proof.

S2. Computational Complexity of Local Pluralistic Homophily

Setting.

Let $G = (V, E)$ with $|V| = N$, $|E| = M$, degree $d_v = |N(v)|$, and a scalar attribute $s : V \to \mathbb{R}$ (as defined in the main text). Recall

$$\tilde{h}_v \;=\; \lambda - \delta_v, \qquad \delta_v = \frac{1}{d_v} \sum_{i\in N(v)} |s(v) - s(i)|, \qquad \lambda = \frac{h + \sum_{u\in V} \delta_u}{N}.$$

Per–node cost (given $s(\cdot)$).

Computing δ_v visits each neighbor once and performs $O(1)$ arithmetic per neighbor; therefore, the cost is $O(d_v)$ time and $O(1)$ extra space.

Global cost (given $s(\cdot)$).

Summing $O(d_v)$ over all nodes yields $O(\sum_v d_v) = O(2M) = O(M)$ time. While accumulating δ_v, we also accumulate $\sum_v \delta_v$ to form λ. A final $O(N)$ pass (or inline) computes $\tilde{h}_v = \lambda - \delta_v$ for all v. Hence, given $s(\cdot)$, the full set $\{\tilde{h}_v\}$ is computable in $O(M)$ time and $O(N)$ space.

End–to–end cost including $s(\cdot)$.

In this work, $s(v)$ counts the number of distinct communities of v present in its neighborhood. Let $\mathcal{C}(v)$ be v's membership set and define the shared–membership multiplicity per edge $\omega(u,v) = |\mathcal{C}(u) \cap \mathcal{C}(v)|$. A direct construction that scans edges and intersects incident membership lists (with hash–set bookkeeping) runs in

$$T_s \;=\; O\Big(\sum_{(u,v)\in E} \omega(u,v)\Big) \quad \text{time, plus} \quad O\Big(\sum_v |\mathcal{C}(v)|\Big)$$

to materialize $s(v)$ from the marks, and an additional $O(M)$ for all δ_v. Thus,

$$T_{\text{end-to-end}} \;=\; O\Big(\sum_{(u,v)\in E} \omega(u,v)\Big) \;+\; O\Big(\sum_v |\mathcal{C}(v)|\Big) \;+\; O(M).$$

Remark. In sparse social/collaboration graphs with moderate overlap, $\omega(u,v)$ is typically small (often 0 or 1), so the practical cost is near–linear in M.

Summary.

Given $s(\cdot)$, the full set $\{\tilde{h}_v\}$ is computable in $O(M)$ time and $O(N)$ space. Including the construction of $s(\cdot)$, an end-to-end bound is $O(\sum_{(u,v)\in E} \omega(u,v)) + O(\sum_v |\mathcal{C}(v)|) + O(M)$.

S3. LFR benchmarks

We use eight LFR graphs with increasing fractions of overlapping nodes (50%–70%). Structural parameters are fixed unless noted: average degree $\langle k \rangle = 20$, maximum degree $k_{\max} = 50$, degree exponent $\gamma = 2.5$, community size exponent $\beta = 1.5$, and overlap per node $O_m = 2$; μ_{mix} denotes the mixing parameter.

Metric divergence via rank correlation.

To quantify how normalization reshapes local rankings, we compute the Spearman rank correlation $\rho(h_v, \tilde{h}_v)$ between the unnormalized h_v and the normalized $\tilde{h}_v$ over all nodes in each LFR network (average ranks for ties). Values near 1 indicate near–identical orderings, whereas smaller values indicate increasing reordering induced by normalization. Table S1 reports LFR configurations together with $\rho(h_v, \tilde{h}_v)$; as the fraction of overlapping nodes grows, ρ declines, evidencing nontrivial divergence under high membership redundancy.

Assortative–disassortative composition under normalization.

Beyond rank divergence, the assortative/disassortative split also shifts with overlap. We label nodes as disassortative if $\tilde{h}_v < 0$ and assortative otherwise. Under higher overlap, the disassortative share shrinks while the assortative share expands, indicating reduced recall but higher selectivity for boundary interfaces (Fig. S1).

Table S1. LFR configuration and rank correlation between the global-centred h_v and the neighbourhood-centred $\tilde{h}_v$. Higher overlap $\Rightarrow$ lower ρ, indicating increasing divergence.

Dataset	N	μ_{mix}	**# Comm.**	**Overlap (%)**	$\boldsymbol{\rho(h_v, \tilde{h}_v)}$
LFR1	5,000	0.10	100	50	0.9985
LFR2	5,000	0.30	100	55	0.9848
LFR3	5,000	0.50	100	60	0.9573
LFR4	5,000	0.70	100	65	0.9178
LFR5	5,000	0.10	100	70	0.8602
LFR6	10,000	0.30	200	50	0.8292
LFR7	10,000	0.50	200	60	0.7924
LFR8	10,000	0.70	200	70	0.7563

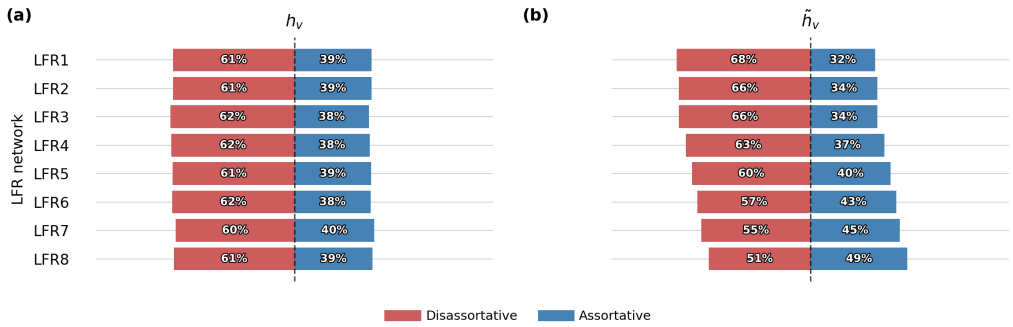

Figure S1. Assortative–disassortative composition across LFR networks with increasing fractions of overlapping nodes (50%–70%). (a) Unnormalized local contribution h_v. (b) Normalized metric $\tilde{h}_v$. Bars show the proportions of disassortative ($\tilde{h}_v < 0$; red) and assortative ($\tilde{h}_v \geq 0$; blue) nodes. As overlap increases (LFR1 $\rightarrow$ LFR8), the disassortative share becomes smaller under $\tilde{h}_v$, consistent with higher selectivity for boundary positions.

S4. Local Pluralistic Homophily vs. Topology-Only Baselines on LFR

Spearman $\rho_s(\tilde{h}_v, \cdot)$ with BridgeCC, degree, and eigenvector remains weak/near-zero across configurations.

Table S2. Spearman $\rho_s(\tilde{h}_v, \cdot)$ on LFR (topology-only baselines).

Dataset	BridgeCC	Degree	Eigenvector
LFR1	−0.132	−0.047	−0.033
LFR2	−0.103	−0.051	−0.033
LFR3	−0.048	−0.020	0.009
LFR4	−0.031	−0.027	−0.007
LFR5	0.008	−0.006	0.029
LFR6	0.029	0.003	0.036
LFR7	0.039	0.003	0.035
LFR8	0.047	0.007	0.037

S5. Community graph and inter–community weight

Edge–community incidence and community graph.

Let $G = (V, E)$ be a (possibly overlapping) community network with community set $\mathcal{C}$ of size C, and let $p \in [0, 1]$ denote the removal budget, with G_p the residual graph. Define the edge–community incidence matrix as:

$$B \in \{0, 1\}^{|E| \times C}, \qquad B_{e\alpha} = \mathbf{1}\big(\mathbf{1}[u \in \alpha] \oplus \mathbf{1}[v \in \alpha]\big),$$

where $e = (u, v) \in E$ and $\oplus$ denotes exclusive OR; thus $B_{e\alpha} = 1$ iff e crosses the boundary of α (exactly one endpoint belongs to α). Let $X(p) \in \{0,1\}^{|E|\times|E|}$ be the diagonal mask indicating whether an edge of G survives in G_p. The (symmetric) community–to–community weight matrix is

$$W(p) = B^\top X(p) B \in \mathbb{R}_{\geq 0}^{C\times C},$$

whose entries quantify boundary mass between community pairs: the diagonal term $W_{\alpha\alpha}(p)$ counts edges incident to the boundary of α, and the off–diagonal term $W_{\alpha\beta}(p)$ counts edges that simultaneously lie on the boundaries of α and β (i.e., inter–community interfaces).

Inter–community weight and normalization.

We summarize inter–community coupling at budget p by the total off–diagonal mass:

$$W_{\text{inter}}(G_p) \;=\; \tfrac{1}{2}\sum_{\alpha\neq\beta} W_{\alpha\beta}(p), \qquad \widehat{W}_{\text{inter}}(p) \;=\; \frac{W_{\text{inter}}(G_p)}{W_{\text{inter}}(G_0)},$$

with $G_0 \equiv G$, so that $\widehat{W}_{\text{inter}}(0) = 1$. The factor $\frac{1}{2}$ avoids double counting due to symmetry.

Basic properties (used implicitly).

(i) Monotonicity: since $X(p)$ is coordinatewise nonincreasing in p, then $W_{\text{inter}}(G_p)$ and $\widehat{W}_{\text{inter}}(p)$ are nonincreasing in p. (ii) Equivalence as off–diagonal mass: $W_{\text{inter}}(G_p)$ equals the sum of the off–diagonal entries of $W(p)$ and can be interpreted as the total boundary mass across community pairs in the community graph induced by G_p.

S6. Edge- and community-level extensions

Edge-level score.

For any edge $e = (u, v)$ and a scalar attribute $s(\cdot)$, define the dyadic dissimilarity

$$\delta_e \;=\; |\, s(u) - s(v)\,|, \qquad M = |E|.$$

Let

$$\lambda_E \;=\; \frac{h + \sum_{e\in E}\delta_e}{M},$$

where h is the network-level assortativity computed over s. The edge-level local pluralistic homophily is

$$\tilde{h}_e \;=\; \lambda_E - \delta_e.$$

Additivity. $\sum_{e\in E}\tilde{h}_e \;=\; M\lambda_E - \sum_e \delta_e \;=\; h$. Thus, the dyadic scores decompose the global coefficient.

Community-level score.

Let $C \subseteq V$ be a community, $E(C)$ its set of internal edges, and

$$\delta_C \;=\; \frac{1}{|E(C)|}\sum_{(u,v)\in E(C)} |\, s(u) - s(v)\,|.$$

Define the community-level local pluralistic homophily

$$\tilde{h}_C \;=\; \lambda_E - \delta_C,$$

which measures within–community alignment relative to the same global baseline λ_E used at the edge level. This choice preserves interpretability (positive values: above-baseline

alignment; negative values: below-baseline alignment) and keeps scores comparable across communities of different sizes and densities.

S7. Empirical Runtime Evaluation of the Local Pluralistic Homophily Score

Table S3 reports wall–clock execution times for computing $\tilde{h}_v$ across the eight real–world networks analyzed in the study. Measurements were obtained on a MacBook Air with Apple M1 chip (8 cores: 4 performance, 4 efficiency), 8 GB RAM, Python 3.12.2, single–threaded execution. Total runtime is decomposed into data loading and preprocessing (T_{load}), computation of the neighborhood–based similarity term $s(v)$ ($T_{s(v)}$; Eq. 3), and computation of the local dissimilarities and scaling factor ($T_{\lambda,\Delta}$; Eqs. 5–6).

Table S3. Wall–clock execution time (seconds) for computing $\tilde{h}_v$ on real–world networks. Components: data loading and preprocessing (T_{load}), neighborhood similarity $s(v)$ ($T_{s(v)}$), and local scores (λ, δ_v) ($T_{\lambda,\delta}$).

Dataset	T_{load}	$T_{s(v)}$	$T_{\lambda,\delta}$	T_{total}
Twitch	0.18	0.06	0.14	0.19
GitHub	0.25	0.52	0.27	0.79
StackOverflow	59.01	975.89	30.71	1006.60
Deezer	0.44	0.50	0.52	1.02
DBLP	1.03	1.47	1.19	2.66
LiveJournal	46.42	198.32	46.23	244.55
YouTube	3.00	3.21	3.45	6.66
Amazon	1.26	1.91	1.07	2.97

Seven of the eight datasets complete in under 250 seconds. StackOverflow requires approximately 1006 seconds ($\approx$ 17 minutes), consistent with its large edge count ($M \approx 63.5$ million) and substantial overlap ($\bar{s} \approx 9.3$). The timing breakdown shows that computation of $s(v)$ dominates in all cases, aligning with the $O(M \cdot \bar{s})$ complexity estimate.

S8. Sensitivity Analysis: Jaccard Threshold τ

Table S4 reports the node-mean inter-community edge ratio (IER) for $\tilde{h}_v$ computed under three Jaccard thresholds, $\tau \in \{0.1, 0.2, 0.3\}$, across eight real-world networks.

Table S4. Node-mean Interface Edge Ratio for $\tilde{h}_v$ across three Jaccard threshold values.

Dataset	$\tau = 0.1$	$\tau = 0.2$	$\tau = 0.3$
DBLP	0.597	0.766	0.837
GitHub	0.090	0.720	0.989
StackOverflow	0.923	0.963	0.979
Deezer	0.280	0.497	0.649
LiveJournal	0.403	0.457	0.472
YouTube	0.046	0.056	0.064
Amazon	0.297	0.408	0.502
Twitch	0.114	0.114	0.114

Across all datasets, IER values increase with τ, reflecting the broader set of edges labeled as inter-community when the threshold becomes less restrictive. The ordering $\text{IER}(\tau{=}0.1) < \text{IER}(\tau{=}0.2) < \text{IER}(\tau{=}0.3)$ holds consistently, and no discontinuities were observed.